

PROGRAMMING FUNDAMENTAL (LAB)

PROJECT

TIC TAC TOE GAME:

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:SOLUTION:

```
#include <stdio.h> // Include the standard input/output library for  
printf and scanf.
```

```
char square[9] = { '0', '1', '2', '3', '4', '5', '6', '7', '8' };
```

```
int checkwin() {  
    if (square[0] == square[1] && square[1] == square[2])  
    {  
        if (square[0] == 'X')  
            return 1;  
    }  
}
```

```

        else
            return 2;
    }
    else if (square[3] == square[4] && square[4] == square[5])
    {
        if (square[3] == 'X')
            return 1;
        else
            return 2;
    }
    else if (square[6] == square[7] && square[7] == square[8])

{
    if (square[6] == 'X')
        return 1;
    else
        return 2;
}
    else if (square[0] == square[3] && square[3] == square[6])

{
    if (square[0] == 'X')
        return 1;
    else
        return 2;
}

```

```
else if (square[1] == square[4] && square[4] == square[7])
```

```
{  
    if (square[1] == 'X')  
        return 1;  
    else  
        return 2;  
}
```

```
else if (square[2] == square[5] && square[5] == square[8])
```

```
{  
    if (square[2] == 'X')  
        return 1;  
    else  
        return 2;  
}
```

```
else if (square[0] == square[4] && square[4] == square[8])
```

```
{  
    if (square[0] == 'X')  
        return 1;  
    else  
        return 2;  
}
```

```
else if (square[2] == square[4] && square[4] == square[6])
```

```
{  
    if (square[2] == 'X')  
  
        return 1;
```

```

        else
            return 2;
    }
    else if (square[0] == square[3] && square[3] == square[6])

    {
        if
        (square[0] == 'X')

            return 1;
    else

        return 2;
    }
    else
        return 0;
}

```

```

void mark(int player, int box) {
    if (player == 1) {
        square[box] = 'X';
    }
    else {
        square[box] = 'O'; // Change 'Y' to 'O'
    }
}

```

```

void display() {
    int i;
    for ( i = 0; i < 9; i++) {

```

```

        printf("%c\t", square[i]); // Use printf for displaying output.
        if (i == 2 || i == 5 || i == 8)
            printf("\n");
    }
}

int main() {
    int player1 = 1, player2 = 2;
    int box, result = 0, flag = 0;
    int i;
    for ( i = 1; i < 5; i++) {
        printf("\nPlayer %d Enter the Box: ", player1);
        scanf_s("%d", &box); // Use scanf for input.
        mark(player1, box);
        display();

        result = checkwin();
        if (result == 1) {
            printf("\nCongratulations! Player %d has Won.\n", player1);
            flag = 1;
            break;
        }
        else if (result == 2) {
            printf("\nCongratulations! Player %d has Won.\n", player2);
            flag = 1;
            break;
        }

        printf("\nPlayer %d Enter the Box: ", player2);
        scanf_s("%d", &box);
    }
}

```

```
mark(player2, box);
display();

result = checkwin();
if (result == 1) {
    printf("\nCongratulations! Player %d has Won.\n", player1);
    flag = 1;
    break;
}
else if (result == 2) {
    printf("\nCongratulations! Player %d has Won.\n", player2);
    flag = 1;
    break;
}
}

if (flag == 0)
    printf("\nSorry the game is a draw.\n");

return 0;
}
```

:OUTPUT:

```
Player 1 Enter the Box: 0
X   1   2
3   4   5
6   7   8

Player 2 Enter the Box: 8
X   1   2
3   4   5
6   7   0

Player 1 Enter the Box: 3
X   1   2
X   4   5
6   7   0

Player 2 Enter the Box: 6
X   1   2
X   4   5
0   7   0

Player 1 Enter the Box: 1
X   X   2
X   4   5
0   7   0

Player 2 Enter the Box: 7
X   X   2
X   4   5
0   0   0

Congratulations! Player 2 has Won.

-----
Process exited after 46.93 seconds with return value 0
Press any key to continue . . .
```

DONE.....